INCH-POUND

MIL-I-38121C (USAF)

30 November 1989
SUPERSEDING
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31 December 1970

MILITARY SPECIFICATION

INDICATOR, PRESSURE EGU-9/A

1. SCOPE

1.1 Scope. This specification covers the requirements for single remote indicating engine pressure ratio indicators, designated EGU-9/A.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

EGU-9/A-1	Nonintegrally Lighted
EGU-9/A-2	Integrally Lighted Red
EGU-9/A-3	Integrally Lighted White

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center /MMEOR, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6620

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issue are those cited in the solicitation.

AF Drawing 8988994 Luminescent Material, Fluorescent.

(Applicaton for copies should be addressed to the OC-ALC/MMIRF, Tinker AFB, OK 73145-5990.)

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-I-7057	Indicator, Synchro, Aircraft, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For

STANDARDS

FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U. S. Military Property
MIL-STD-470	Maintainability Program Requirements (For Systems And Equipments)
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Testing For Engineering Development, Qualification, & Production
MIL-STD-785	Reliability Program For Systems And Equipment Development And Production
MIL-STD-810C	Environmental Test Method And Engineering Guidelines
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards And Specifications, Order Of Preference For The Selection Of
MIL-STD-2073/1A	DOD Material Procedures For Development And Application Of Packaging Requirements
MS33558	Numerals And Letters, Aircraft Instrument Dial, Standard Form Of
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Unless otherwise indicated, copies of federal and military specifications,

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standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society For Testing and Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualifications. Indicators furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.4 and 6.3).

3.2 General specification. The requirement of MIL-I-7057 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Selection of specification and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.4 Materials

3.4.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service.

3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4.2 Protective treatment. Materials that are utilized within a hermetically sealed case shall be considered suitably protected against corrosion. Aluminum-alloy parts shall be coated with a anodic film conforming to MIL-A-8625 when used outside the hermetically sealed inclosure.

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TABLE I. Indicator Pressure Ratio Tolerances.

Pressure Ratio	Pressure Ratio Tolerance (±)		
	High Temperature	Low Temperature	Room Temperature
1.1	0.005	0.005	0.004
1.5	0.005	0.005	0.004
1.7	0.005	0.005	0.004
2.0	0.005	0.005	0.004
2.3	0.005	0.005	0.004
2.5	0.005	0.005	0.004

3.5 Design and construction. The indicator shall be of a synchro design for use with a pressure ratio transmitter Honeywell part number LG1197AA03 to indicate at a remote station the pressure ratio units of turbofan engine operation.

3.5.1 Calibration. The indicator dial shall have linear graduations in accordance with figure 1. The calibration of pressure ratio units versus angular degrees shall be linear within ± 10° 15' of arc.

3.5.2 Integral lighting. The indicator shall be designed for integral lighting in accordance with MIL-L-25467 or MIL-L-27160, as applicable.

3.5.3 Reliability. Engineering for Reliability shall occur simultaneously with engineering to achieve electrical and mechanical characteristics. The equipment lower test mean-time-between-failure (MTBF), shall not be less than 4000 hours. Failure refers to any condition of the equipment whereby full operational service, as required herein, is not possible.

3.5.4 Maintainability. The indicator shall be maintainable within the general parameters of MIL-STD-470.

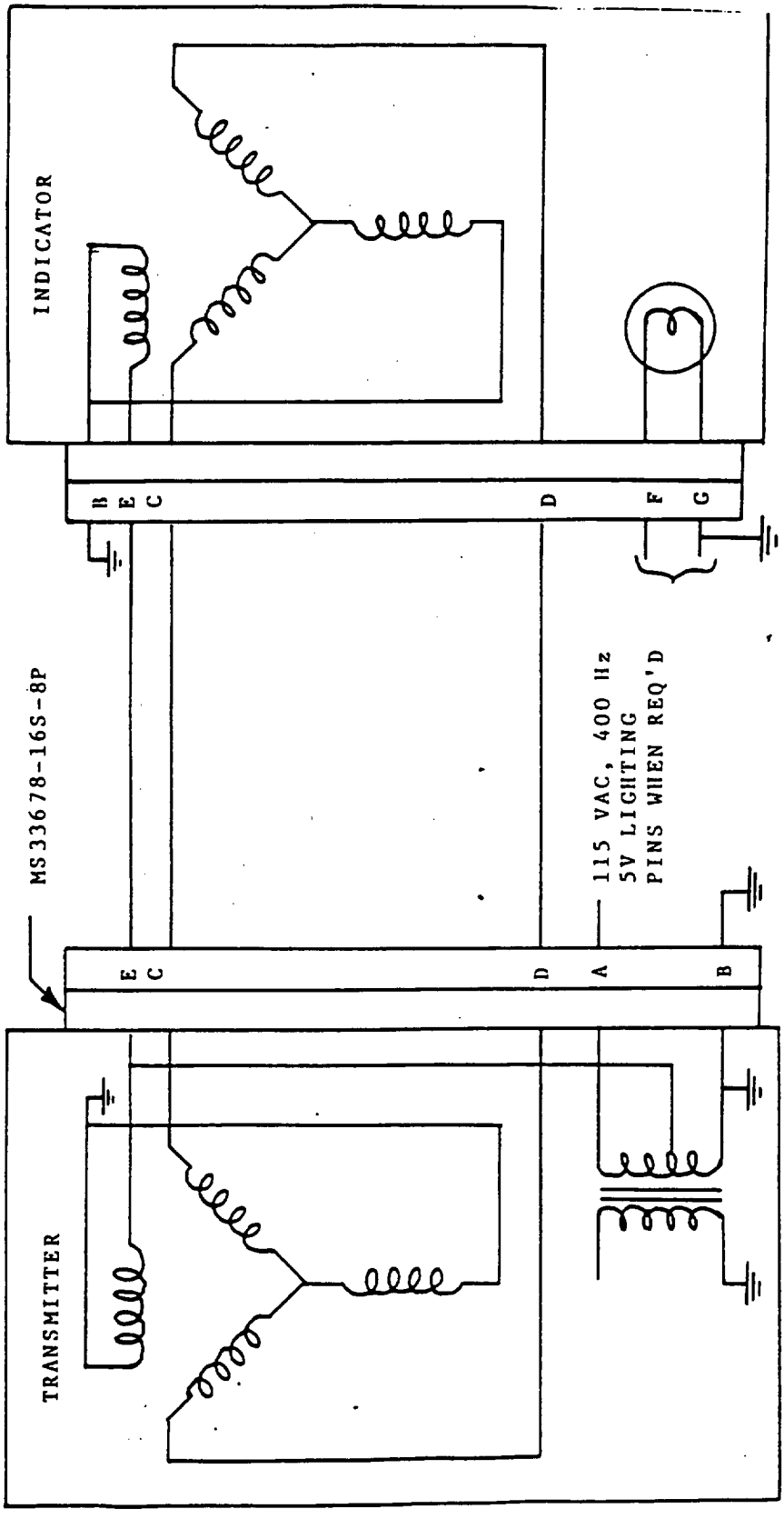
3.6 Performance. The performance of the indicator shall be within the tolerances specified in tables I and II when connected to a master transmitter energized at 26 ± 0.05V and 400 ± 1 cycle single-phase power. System wiring as illustrated in schematic 1 shall cause CW indicator pointer rotation.

TABLE II, Performance Tolerances.

Test	Pressure Ratio Tolerance (±)
Frequency and Voltage Variation	0.005
Position Error	0.005
Friction:	
Room Temperature	0.005
Low Temperature	0.010
High Temperature	0.010
Vibration:	
Pointer Oscillation	0.010
Pointer Variation	0.010
Counter and Pointer Index Cycling	0.010

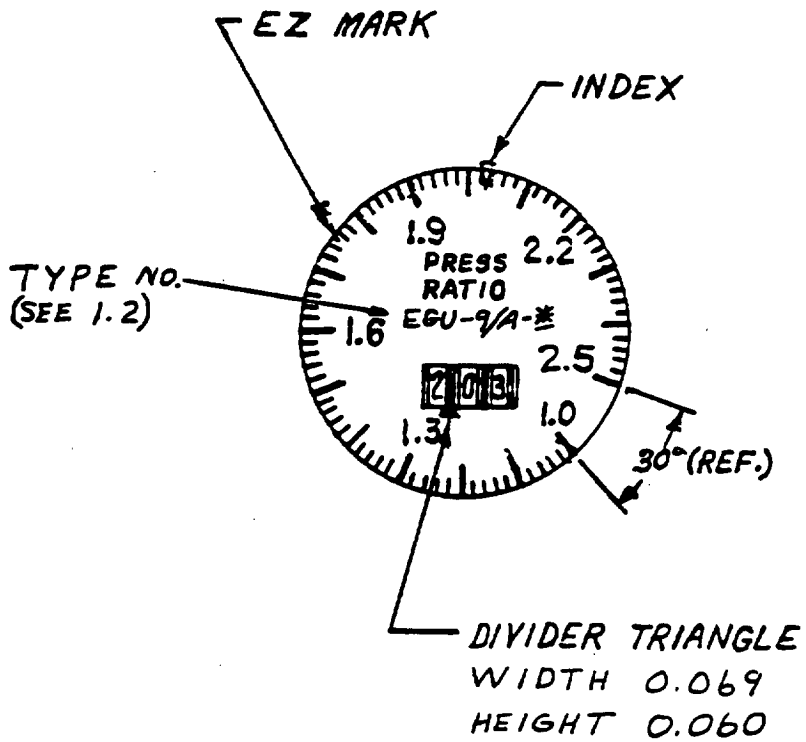
3.6.1 The indicator shall be capable of meeting the test requirements specified herein when subjected to the following conditions:

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Scematic 1. Wiring diagram.

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NOTES: 1.6 EPR MARK LOCATED AT 9 O'CLOCK DIAL POSITION THE DIVIDER TRIANGLE SHALL FORM PART OF THE DIAL.

FIGURE 1. Dial.

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a. Operating temperatures ranging from -55° to +85° C and exposure to temperatures ranging from -65° to +85°C.

b. Vibration incident to service use.

c. Salt-laden atmosphere.

d. High altitude and lower temperature, as specified.

3.6.2 Magnetic effect. When the indicator is held 5.50 inches from the center of a short bar magnetic compass and rotated in a vertical plane, the compass shall not deflect more than 3°.

3.7 Case. The case shall conform to figure 2 and shall be hermetically sealed. The length of the case, excluding the connector, shall be 5 inches maximum. It shall be finished in lusterless black, color No. 37038 of FED-STD-595.

3.7.1 Filling medium. The case shall be filled with 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dew-point -65°C) at the filling pressure. The absolute pressure of the filling medium shall be 1 atmosphere ± 1 psi.

3.7.2 Filling tube. The filling tube, if used, shall be as least 1.25 inches long and shall not be visible when the indicator is completely assembled.

3.8 Dial. The dial shall be securely attached to the frame of the indicator mechanism and shall conform to figure 1.

3.8.1 Dial diameter. The dial diameter shall be the maximum possible consistent with good design practice. Large gaps between the edge of the dial and the indicator case will not be acceptable.

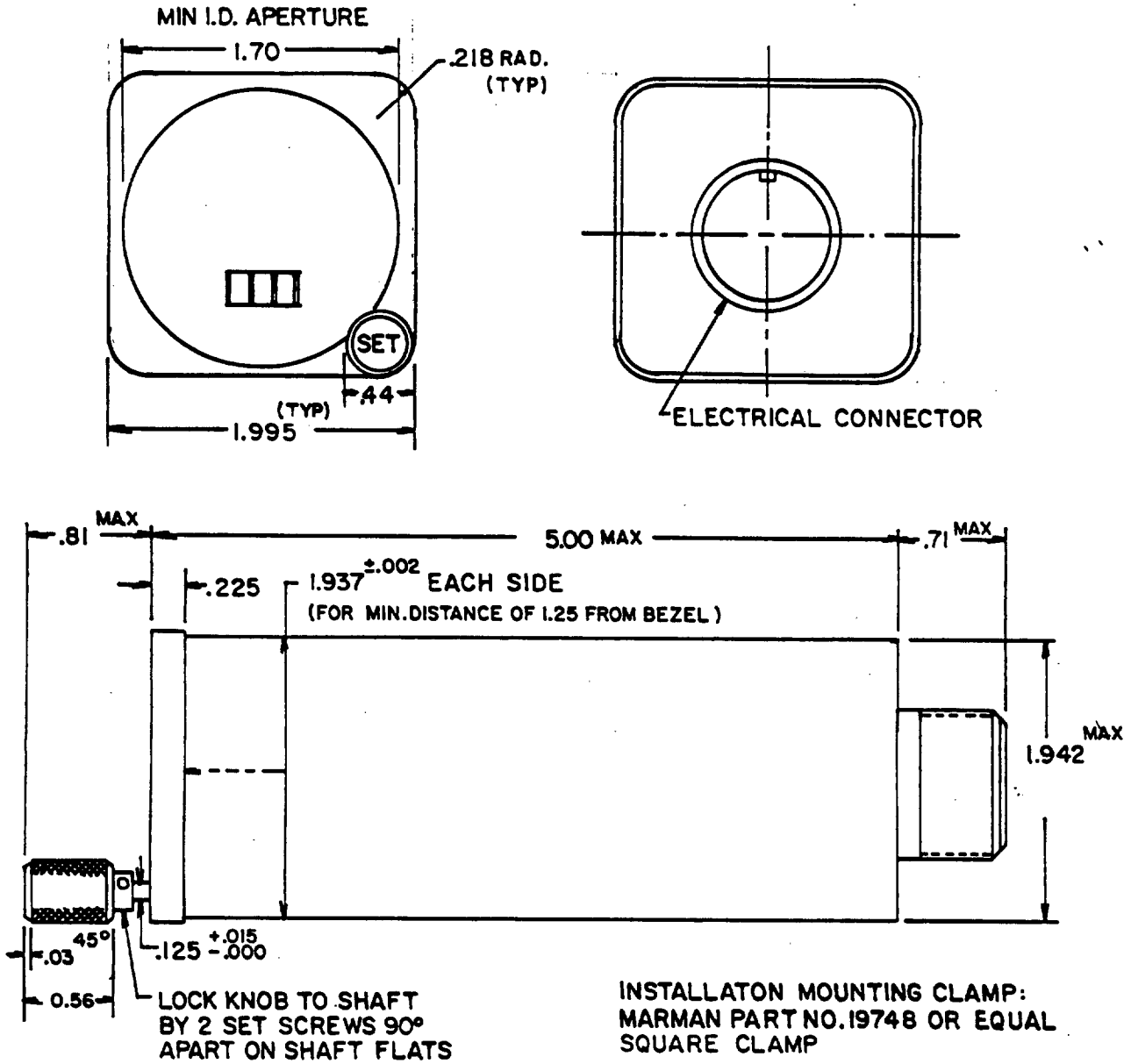
3.8.2 Dial to cover distance. The distance between the indicator dial and the inside of the glass nearest the dial shall not exceed 0.188 inch.

3.8.3 Dial markings.

3.8.3.1 Finish and dimensions. The dial markings for nonlighted indicators shall conform to MS33558 and be finished in a light yellow fluorescent luminescent paint in accordance with AF drawing 8988994. For integrally lighted indicators, the dial markings shall be finished in white, color No. 37875 of FED-STD-595. The dimensions shall be as follows:

	Height or Length Inch + 0.16	Width of Linear Graduation Inch +0.005
Scale numerals 1.0 through 2.5	.156	0.031
0.10 Graduations	.187	0.031
0.02 Graduations	.109	0.020
Lettering PRESS RATIO	.094	0.016
Electrical zero reference mark	.031	0.015
Counter numerals	.125	0.020
Knob lettering SET	.125	0.020

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DMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES: DECIMALS ±.005

FIGURE 2. Case.

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3.8.4 Pointer. The pointer shall conform essentially to MS33585-8 and shall be suitable for either integral or nonintergral lighting, as applicable. Pointer hub or tail shall not overlap the counter readout at any position. A black index, 0.016 inch wide by 0.220 inch long, shall be provided to ease alignment of the pointer to the settable index. It shall be located on the centerline of the pointer.

3.8.5 Takeoff index and counter. The dial shall contain a moveable index and a drum-type counter for takeoff-pressure-ratio-reference indication. The index shall conform to figure 1. The index shall be mechanically related to the counter marked in pressure ratio shown on figure 1. The maximum allowable position error between the counter and the point of the index shall be ±0.005 pressure ratio units. The counter shall indicate pressure ratio from 1.0 to 2.5. The index and associated counter shall move smoothly over the complete scale range of 1.0 to 2.5 pressure ratio when a torque of 6 ± 3 ounce-inches is applied to the control knob. The index and counter shall be moved by the indexing mechanism knob when the knob is turned. The length of the index shall be 0.203 ± The counter numerals shall move downward with an increasing pressure ratio as the knob is rotated cw. The index mechanism employed for driving the index and counter over its design range shall be rugged in design. The control knob and drive shaft shall conform to the dimensions specified on figure 2. There shall be no binding, deformation, or malfunction when the indexing mechanism is rotated over its design range. Suitable rugged stops or slip clutch mechanism shall be provided to keep the index and counter from being damaged or mismatched if rotated above or below the dial range.

3.8.6 Knob. The knob shall conform to figure 2, and shall be finished in durable lusterless black, color No. 37038 of FED-STD-595. The lettering shall be finished in durable dull white, color No. 37875 of FED-STD-595, and shall be as large as possible. The sides of the knob shall be knurled.

3.9 Electrical receptacle. The indicator shall be provided with an MS33678H-16S-1P electrical receptacle suitable for hermetic sealing. Pin connections shall be as follows:

Pin	
A	Spare
B	Ground
C	Stator
D	Stator
E	Power (26V 400 Hz)
F	5V Light (when lighted)
G	5V light ground

3.10 Power. The indicator shall be designed to operate from 26V ± 4V, 400 Hz ± 20 percent single-phase a-c power in accordance with the characteristics specified in MIL-STD-704 except that the ratio of volt to frequency shall be 26V/400 Hz, ± 5 percent. The input power to the indicator shall be not more than 6 va excluding lighting.

3.11 Weight. The weight of the indicator shall not exceed 0.85 pound.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following identification shall be included on the indicator nameplate:

Indicator, Pressure EGU-9/A-_____*
*Insert applicable type designation.

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3.13 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in this specification.

4.3.1 Atmospheric. Unless otherwise specified, all tests shall be conducted at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressures or temperatures substantially different from the above values, proper allowances shall be made from the specified condition.

4.4 Qualification inspection. The qualification inspections identified in Table III shall be performed as described under 4.6.

4.4.1 Test samples. The test samples shall consist of a minimum of seven indicators representative of the production equipment. Samples shall be identified with the manufacturer's part number and such other information as required by the contracting activity.

4.5 Quality conformance inspection. The quality conformance inspections identified in Table III shall be performed as described under paragraph 4.6 and consists of the following:

a. Individual inspections
b. Sampling plans and tests

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4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6.

a. Examination of product
b. Electrical zero
c. Scale error and friction
d. Position error
e. Integral lighting, when applicable
f. Dielectric strength

TABLE III, Examinations and Tests.

Test	Paragraph Number	Qualification Test	Quality Conformance Test
Examination of Product	4.6.1	X	X
Electrical zero	4.6.2	X	X
Scale error and friction	4.6.3	X	X
Vibration	4.6.4	X	X
Low temperature	4.6.5	X	X
High temperature	4.6.6	X	X
Magnetic effect	4.6.7	X	X
Voltage and frequency variation	4.6.8	X	X
Fogging	4.6.9	X	
Sealing	4.6.10	X	
Extreme low temperature exposure	4.6.11	X	
Extreme high temperature exposure	4.6.12	X	
Counter and pointer index cycling	4.6.13	X	X
Salt Fog	4.6.14	X	
Reliability	4.6.15	X	X
Dielectric strength	4.6.16	X	X
Torque	4.6.17	X	X
Lead length	4.6.18	X	X
Thermal shock	4.6.19	X	X
High altitude-low temperature	4.6.20	X	X

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. Three indicators shall be selected at random from each 100 or fraction thereof produced, and shall be subjected to the following tests as described in 4.6.

a. Vibration
b. Low temperature
c. High temperature
d. Magnetic effect
e. Voltage and frequency variation
f. Torque
g. Lead length
h. Sealing
i. Fogging

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4.5.2.2 Sampling plan B. Unless otherwise specified, 3 indicators shall be selected at random from the first 10 items produced, and shall be subjected to the following tests as described under 4.6.

a. Counter and pointer index cycling
b. Thermal Shock
c. High altitude - low temperature
d. Extreme low temperature exposure
e. Extreme high temperature exposure
f. Salt fog
g. Reliability

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and methods of correcting them.

4.6 Test Methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 Electrical zero. The electrical zero test and the verification of leads shall be accomplished on the completed indicator with the case removed. A voltage of 26 $\pm$ 1 V single-phase shall be applied to the rotor of the synchro with leads designated R_1 and R_2. R_2 shall be connected to the ground or low potential side of the power input. With a lead designated S_2 and connected to R_2, the following conditions shall exist at electrical zero:

a. The voltage between the remaining stator terminals designated S_1 and S_3 shall be at minimum.

b. The voltage between R_1 and S_3 shall be near the low null and shall reach the low null when the rotor is rotated cw approximately 30° from electrical zero.

c. Clockwise rotation of the rotor not exceeding 30° from electrical zero shall cause an increase in the voltage between S_3 and S_2 and a decrease in the voltage between S_2 and S_1.

d. The voltage $E(S_3S_2)$ shall be approximately in phase with the rotor voltage $E(R_2R_1)$. The sequence of the terminal designations in parentheses indicates the direction of the voltage vector.

e. When the rotor is turned counterclockwise (ccw) not exceeding 180° from the

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electrical zero, the voltage $E(S_1S_3)$ shall remain approximately in phase with the rotor voltage $E(R_2R_1)$. At electrical zero, the tip of the pointer shall be within 0.010 inch of the electrical zero reference mark.

4.6.3 Scale error and friction. During the scale error test, the indicator can be connected to a master transmitter especially designed for this type of testing. The master transmitter shall not have an error of more that 0.05 circular degrees and shall be furnished with a correction table so that the correct transmitter position can be determined within 0.05 circular degrees. The corrected master transmitter readings required to provide the respective test indicator readings listed in table I shall be set while the indicator is being gently tapped or vibrated. The indicator shall be considered out of tolerance when the difference between the indicator reading and master synchro transmitter reading exceeds the tolerance shown in table I. The indicator friction error shall be measured by setting the indicator on the respective test points shown in table I while it is gently tapped or vibrated. At each test point and without tapping or vibrating the indicator, the master synchro transmitter shall be advanced to an angular position which is just sufficient to cause the indicator pointer to move. The indicator dial shall also be scanned and observed for points of excessive friction at intermediate points. Friction error shall be considered as the difference between the master synchro transmitter setting required to produce the indicator setting while under vibration or tapping and the master syncho transmitter setting required to move the pointer without vibration or tapping.

4.6.4 Vibration. The indicator shall be subjected to a vibration test in accordance with method 514.2, equipment category b.1, procedure I (part 1 only), curve B of MIL-STD-810C. No malfunction of the indicator shall result from this test. Pointer oscillation and variation shall be within the tolerances specified in table II.

4.6.4.1 Scale error after vibration. Following the vibration test, the indicator shall be checked for scale error and friction. The room temperature tolerances specified in table I and II shall apply.

4.6.5 Low temperature. The indicator shall be placed in a test chamber and subjected to $-55° \pm 2°$ C for a period of 4 hours. At the end of this period and while still at the specified temperature, the indicator shall be subjected to the test specified in 4.6.3. The scale errors shown by the indicators shall not exceed the tolerances specified in tables I and II. The indicator shall be returned to room temperature for a minimum period of 4 hours and again subjected to the scale error at room temperature test. The scale error shall not exceed the tolerance specified in table I.

4.6.6 High temperature. The indicator shall be placed in a test chamber and subjected to a temperature of $71° \pm 2°$ C for a period of 4 hours. At the end of this period and while still at this temperature, the indicator shall be subjected to the test specified in 4.6.3. The scale errors shall not exceed the tolerances specified in tables I and II. The indicator shall be returned to room temperature for a minimum period of 4 hours and again subjected to the scale error at room temperature test. The scale error shall not exceed the tolerance specified in table I.

4.6.7 Magnetic effect. The indicator, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnetic compass with the nearest part of the indicator 5 ± 0.50 inch from the magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in

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position 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated about its own horizontal axis until it is in its normal viewing position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 3°.

4.6.8 Voltage and frequency variation. The indicator shall be subjected to the test specified in 4.6.3 and the input power varied to provide any combination of the following voltages and frequencies. The indicator scale errors at the varied voltages and frequencies shall not exceed the tolerances specified in table II.

a. 22 to 30V

b. 380 to 420Hz

4.6.9 Fogging. The indicator shall be energized and in a controlled ambient of 70 ± 2°C for minimum of 1 hour. At the end of this period and while at the specified temperature, an ice cube made of distilled water and of nominal original size, 1 by 1 inch, with its flat surface 1 by 1 inch, shall be rubbed over a major area of the indicator glass face from 1 to 2 minutes. If the ice cube melts to an unusable size before this test is completed. A second ice cube shall be used. The glass shall be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of internal water or oil fog. Evidence of fogging shall be cause for rejection.

4.6.10 Sealing. The hermetically sealed indicator shall be tested for leaks by means of a mass-spectrometer type helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 1.0 micron cubic foot per hour.

4.6.11 Extreme low temperature exposure. The indicator shall be placed in a test chamber and subjected to a temperature of -65°± 2°C for a period of 48 hours. While at this temperature, no operational test shall be performed. The temperature shall be raised to -54°±2°C and maintained for 24 hours. While still at -54° ± 2° C, the indicator shall be subjected to the error test at the test points specified in table I. A scale error tolerance of ± 0.005 EPR shall apply. The indicator shall be returned to room temperature for a minimum period of 4 hours and subjected to the test specified in 4.6.3.

4.6.12 Extreme high temperature exposure. The indicator shall be placed in a chamber and subjected to a temperature of 85° ± 2° C for a period of 24 hours. Normal instrumentation power shall be applied continuously the last 2 hours of this period. While still at the specified temperature, the indicator shall be operated throughout its range and shall exhibit no sticking or erratic characteristics. The temperature shall then be lowered to 71° ± 2°C for a 4 hour period and the indicator subjected to the high temperature test; the tolerances specified in tables I and II shall apply. The indicator shall be returned to room temperature for a minimum period of 2 hours, then subjected to the scale error at room temperature test. The scale errors shall not exceed the tolerances specified in table I.

4.6.13 Counter and index pointer cycling. The set knob shall be turned continuously in order to move the takeoff index and counter from 1.1 to 2.3 ± 0.1 and return to 1.1 ± 0.1 pressure ratio units. This procedure shall be considered one cycle of operation. The total cycling period shall be a minimum of 2,500 cycles

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and shall be conducted at a cycling rate of 2 ± 0.2 Hz. Prior to the cycling test, the torque required to turn the control knob shall be measured and recorded. The allowable torque required to turn the control knob throughout the range of 1.0 to 2.5 pressure ratio units shall not exceed 6 ± 3 ounce-inches.

4.6.13.1 Upon completion of the 2,500-cycle testing, physical and functional operation inspection shall be conducted to assure that binding, deformation, and malfunction are not evident. The torque required to turn the control knob throughout the range of 1.0 to 2.5 pressure ratio units shall not exceed the measurements recorded at the start of the cycling test by more than 2 ounce-inches.

4.6.14 Salt fog. The indicator shall be subjected to a salt fog test in accordance with method 509.1, procedure I of MIL-STD-810C for a 50-hour period. The scale error at room temperature test shall be conducted immediately before and after this test. The scale error tolerances shall not exceed those specified for room temperature. Subsequent indicator operation shall not be affected by this test, and excessive corrosion of metal parts, including electrical receptacles, shall be cause for rejection.

4.6.15 Reliability. The indicator shall be subjected to a reliability demonstration at the following conditions of temperature, vibration, and input voltage. The cycling of these conditions shall be applied to the equipment during the test:

a. Temperature: -55° C to +85° C (-67° F to +185° F)

b. Temperature cycling: Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature, plus two hours. Stabilize the equipment temperature at the lower test level temperature. Turn the equipment on and allow it to experience the combined effect of chamber heating and equipment operation until the point of maximum thermal inertia stabilizes at the higher temperature, that is, when the rate of thermal change is less than 2° C/hr. Maintain the equipment at the upper test level temperature for a period of two hours. Turn the equipment off and allow it to experience chamber cooling until the point of maximum thermal inertia stabilizes at the lower test level temperature. This cycle shall continue for the duration of the test.

c. Vibration: The equipment shall be subjected to a swept sinusoidal vibration conforming to curve AR of Fig. 514 2-2 of MIL-STD-810C, Method 514.2. The equipment shall be swept from 5 - 2000 Hz within 15 minutes. Vibration shall be applied 15 minutes during each hour of equipment operating time. When the equipment being tested contains circuit boards or cards, vibration shall be normal to the plane of the majority of the cards. Otherwise, the direction of vibration is not critical. This vibration requirement may be met by mounting the equipment solidly, (that is, without the equipment shock vibration mounts) to a strong flat plate which is supported by vibration isolators. The desired amplitude and frequency may be obtained by a simple vibration machine with an asymmetric weight on a shaft attached to the plate and driven by a suitable motor. It is recommended that the vibration equipment be checked for proper operation each 24 hours of operation, and that the vibration transducer be "on" and monitored continuously during vibration.

d. Equipment on-off cycling: Equipment off during cooling cycle and on during heating cycle.

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e. Input voltage: Nominal specified voltage +5 -2%.

f. Input voltage cycling: Input voltage shall be maintained at 110 percent of nominal during the first third of each on cycle, at nominal for the second third of each on cycle, and at 90 percent of nominal during the last third of each on cycle.

At least once every 100 hours of testing, the indicators shall be subjected to a scale error and friction error test at room temperature. A failure shall be counted for each indicator that fails to pass these tests. The accept-reject criteria shall be that of Test Plan I of MIL-STD-781. The lower test MTBF (0) shall not be less than 4000 hours. The equipment shall be operated through the entire performance profile specified herein during the Reliability Demonstration Test. No preventative maintenance shall be performed during the reliability tests.

4.6.15.1 Report. Within 90 days after receipt of a contract and 30 days prior to the scheduled date of initiation of testing, or at such later date as the contracting officer may authorize, the contractor shall submit a detailed reliability test program in accordance with MIL-STD-785 to the contracting activity for approval. Testing shall begin only after formal approval of the test program by the contracting activity.

4.6.16 Dielectric strength. A voltage of 500 a-c rms at commercial frequency shall be applied between each pin of the electrical receptacle and the indicator case for a minimum period of 5 seconds. There shall be no insulation breakdown. If the indicator case is filled with an inert helium gas and hermetically sealed, the potential shall be reduced to 220V and the same test conducted.

4.6.17 Torque. The indicator shall be connected to the master transmitter synchro which shall be locked in the electrical zero position. The synchro rotor shall then be energized with rated voltage and frequency $\pm$ 1 percent. Weights shall be applied to a torque pully on the rotor shaft which shall cause the shaft to rotate in a ccw direction through an angle of 5° of arc. The torque tending to return the shaft to the original position shall be computed and shall not be less than 2,000 milligram/millimeter per degree of displacement.

4.6.18 Lead Length. The indicator shall not change more than 1° of arc when the lead length of No. 20 AWG gage wire between the indicator and transmitter is changed from 5 to 500 feet. Equivalent resistances may be substituted for the added wire length. The ground lead to pin A shall be kept short.

4.6.19 Thermal shock. The hermetically sealed indicator shall be subjected to 8 cycles of immersion in water at a temperature of 85° $\pm$ 2° C and then in water at a temperature of 5° $\pm$ 2°C. There shall be no evidence of moisture penetration or damage to the coating or the case. The length of time for each immersion shall be 30 minutes with not more than 5 seconds between immersions. After 8 continuous cycles have been completed, the indicator shall be subjected to the test specified in 4.6.10. Any evidence of leakage or fogging of the cover glass shall be cause for rejection.

4.6.20 High altitude-low temperature. The indicator shall be installed in a temperature, altitude chamber equal to MIL-STD-810C in a manner similar to its normal mounting on an aircraft instrument panel. The high-altitude low-temperature test shall be conducted at a temperature of -55° C and with a chamber pressure equivalent to an altitude of 50,000 feet for a period of 4 hours. While under these conditions and at the end of the 4-hour period, a scale and friction error test shall be conducted. The scale errors shall not exceed the tolerances specified in tables I and II.

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4.7 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Indicators shall be cleaned in accordance with process IC-I of MIL-P-116.

5.1.1.2 Drying. Indicators shall be dried in accordance with process IC-1 of MIL-P-116.

5.1.1.3 Preservation application. Preservation shall not be used.

5.1.1.4 Unit Packaging. Unless otherwise specified by the contracting activity, each Indicator shall be packaged in quantity unit packs of one each in accordance with Method of IC-I MIL-P-116. Each Indicator shall be placed in a PPP-B-636 fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the Indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each Indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of the Indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged Indicator shall be packed in accordance with ASTM D3951.

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5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The EGU-9/A indicators covered by this specification are intended for use in indicating aircraft turbofan engine pressure ratio.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Type of indicator required

c. When sampling plan B tests will not be conducted

d. Applicable level of packaging

e. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Definitions.

6.3.1 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets or "O" rings.

6.3.2 Amplitude. Amplitude is the extent of motion as measured from one extreme to the opposite extreme.

6.4 Qualifications. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for Qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified List is Oklahoma City Air Logistics Center/MMEOR, Tinker AFB, OK 73145-5990, and information pertaining to qualification of products may be obtained from that activity.

6.5 Subject term (keyword) listing.

Engine Pressure Ratio
1 to 2.5 Indication

6.6 Changes from previous issue. Marginal notations are not used in the revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE - 99

Preparing Activity:
AIR FORCE - 71

Project Number:
6620-F476

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-3812/1C(USAF)	INDICATOR, PRESSURE EGU-9/A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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